

CEG 2400 – Network Installation Proposal Project
100 Points - Due NO LATER than 11:59PM
WEDNESDAY, APRIL 23, 2025

****ALL SUBMISSIONS MUST BE MADE IN PILOT TERM PROJECT DROPBOX****

Grading Criteria and Project components

This final project is intended to help in building the skills needed to read, evaluate and respond to a scope of work outline or a request for proposal for small / medium business grade network. Final projects will be graded on the scale outlined below.

1. One Page Summary: Prepare a one-page summary that outlines your approach to the network design in narrative format.

Professional appearance / formatting - 5 points

Quality of narrative - 5 points

Length very close to but **not exceeding** 1 page with 1 inch margins and 12 point type face - 5 points

Technical relevance - 5 points

2. Network Diagram: Provide a network diagram that depicts the physical layout of the proposed networking solution.

Quality of the diagram - 10 points

Technical accuracy of the diagram - 10 points

3. Price list and Parts summary: Prepare an EXECUTIVE summary of all parts and pricing in spreadsheet format. This summary should include the item, quantity, description including model number if applicable and pricing. The sheet should also include a total cost for the project.

Professional appearance of the data - 10 points

Technical accuracy of the items listed - 10 points

4. Network Tree / User Diagram: Prepare a proposed diagram depicting the construction of the network tree that will contain users, servers, printers and other specified resources. This should be provided in hierarchical tree structure.

Professional appearance / quality of the tree diagram - 10 points

Technical accuracy of the diagram - 10 points

5. PowerPoint Presentation: Prepare a PowerPoint slide deck that contains all of the above items. The slide deck should include the following. EVERY presentation should include a title slide with your name, the date and the class information. The slide deck should have a professional appearance.

Summary Slide - containing no more than 5 bullet points outlining your proposal - 5 points

Network Diagram Slide - 5 points

Price list summary slide - 5 points

Network Tree / Diagram Slide - 5 points

Request for Proposal

Summary

Techopoly Inc. is relocating to a new headquarters in Dayton, OH. Techopoly has obtained an existing single level 50,000 sq. ft. commercial structure that is serviced by a full dedicated T3. This T3 is fully terminated in the facility. Techopoly will be building out the interior of the structure in accordance with architectural recommendation and the infrastructure recommendations from this bid submissions.

Bidders will be required to provide a physical network diagram, costing summary with item descriptions, Hierarchical network resource diagram and scope of work summary with accompanying PowerPoint presentation.

Physical Network

Techopoly is seeking the procurement and installation of a network meeting the following criteria:

Items marked with an * MUST be represented in the network resource diagram.

- *48 port network SWITCH (managed or unmanaged)

- *4 port Fiber Optic Switch

- 3 Multimode fiber optic cables 6 - 10 foot

- 3 Fiber Optic NIC cards (PCI or PCI-e)

- CAT 7 - Unshielded Twisted Pair Cabling Estimate based on an AVERAGE run length of 50 feet.

- "\$5,000 Cable Termination and Installation Allotment" – (THIS ITEM SHOULD BE LISTED IN YOUR PROPOSAL).

- *1 Small / Medium business grade firewall appliance.

- *3 Enterprise grade servers with DEDICATED UPS power protection. (THESE WILL ACCOUNT FOR 3 of 5 UPS)

- *30 Microsoft Windows 11 PRO complete workstations. (Monitor, CPU, Mouse and Keyboard)

- *5 medium duty NETWORKED printers (i.e., Hp LaserJet M507X)

- 35 adequate power strips

- 5 2KVA / 2200VA or higher SMART UPS. (3 of these are for the SERVERS)

Network Resource Layout

Bidders will also provide a layout of the resources to be placed into a Microsoft Active Directory tree structure. This diagram should be designed to minimize administrative overhead and use AD organizational units and groups to isolate and secure objects in the tree. Please reference the attached organizational chart and the above physical network resources that will be represented as objects in the tree.

